MODULE 20 · DOMAIN 3 - APPLICATION DEVELOPMENT

Evaluation vs monitoring, and choosing chunking from evidence

Testing before release, watching after release, and letting results pick the chunking strategy.

SECTION 3 of 6 EXAM WEIGHT 30% MODULE 20 of 40

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Evaluation is the driving test. Monitoring is the dashcam. One proves you can drive before you go out. The other shows how you actually drive once you are out there.

Evaluation or monitoring?

EVALUATION

Before release, offline

- A fixed evaluation dataset
- Expected answers are available
- Compare versions side by side
- mlflow.genai.evaluate

MONITORING

After release, online

- Live production traffic, sampled
- No expected answers
- Spot drift, failures, new patterns
- Registered scorers on traces

THE TIEBREAKER

Deciding whether a change is better is evaluation. Finding out how the live app is doing is monitoring.

What gets checked at each phase

01
Develop
Quick evaluations while iterating.

02
Pre-release
Full evaluation set as a gate.

03
Release
Deploy the version that passed.

04
Production
Sampled monitoring on live traces.

05
Feed back
Failures become new evaluation rows.

ON THE EXAM

The loop closes: production failures join the evaluation set, so the next release is tested on them.

Production monitoring with MLflow 3

ASPECT	DETAIL
How	Register a scorer, then start it with a ScorerSamplingConfig
Sampling	sample_rate trades coverage against cost; filter_string targets traces
Results	Attached as feedback to each evaluated trace
Consistency	The same scorers run in development and production
Limits	At most 20 scorers per experiment. Feature is in Beta

ON THE EXAM

Expensive judges can run on a small sample, and cheap code-based checks on everything.

Which judges can run on live traffic?

NEED EXPECTED ANSWERS

Evaluation only

- Correctness
- RetrievalSufficiency
- Anything using expected facts
- Live requests have no answer key

NEED NO EXPECTED ANSWERS

Evaluation and monitoring

- Safety and Guidelines
- RelevanceToQuery
- RetrievalGroundedness and RetrievalRelevance
- Code-based checks, such as length

THE TIEBREAKER

A judge that needs an answer key cannot mark live traffic, because live traffic has no answer key.

Signals that only production can give you

- **Real questions**, including ones nobody thought to test.
- **Drift**: quality slowly falling as data or users change.
- **Multi-turn** judges over whole conversations, grouped by session.
- Operational signals: latency, errors, cost and token use. Module 34 covers them.
- Human feedback from users and experts. Module 32 covers it.

> ON THE EXAM
>
> Evaluation can only test what you thought of. Monitoring shows you what you did not think of.

Let the results pick the strategy

01
Fix the set
Same questions, same expected facts.

02
Build variants
Sizes, overlaps, or strategies.

03
Score retrieval
Recall and RetrievalSufficiency.

04
Score cost
Records, latency, tokens per request.

05
Choose
Passes the bar at the lowest cost.

ON THE EXAM

The same rule as module 12: apply the quality threshold first, then take the cheapest option that passes.

Four chunk sizes, one requirement

CHUNK SIZE	RECALL AT 5	SUFFICIENCY	RECORDS
256 tokens	0.71	68%	4.1 million
512 tokens	0.84	81%	2.2 million
1,024 tokens	0.86	83%	1.2 million
2,048 tokens	0.79	74%	0.6 million
Requirement	-	at least 80%	fewest that pass

ON THE EXAM

Quality rises then falls as chunks grow. Only 512 and 1,024 pass, and 1,024 needs far fewer records.

Harbourline after launch

01
Gate
300 questions, Correctness at least 90%.

02
Monitor
Safety on all traces, Guidelines on 20%.

03
Spot
Guidelines failures rise on pet cover.

04
Feed back
Failing traces added to the evaluation set.

05
Fix
New release must pass the bigger gate.

ON THE EXAM

Correctness gated the release. It could not monitor live traffic, so Guidelines and Safety did that job.

> A team has rewritten its assistant's system prompt. Before releasing it, they want to know whether it is better than the current one.
>
> They have 300 curated questions with expected answers.

What should the team do?

A Run mlflow.genai.evaluate on the dataset for both prompt versions

B Register a production monitoring scorer on live traffic

C Watch the latency chart on the serving endpoint after the release

D Release it and ask users to rate the new version afterwards

◆ PAUSE AND DECIDE

What should the team do?

A Evaluate both versions

Before release, same dataset, expected answers: offline evaluation.

B Production monitoring

Watches live traffic. The new prompt is not live yet.

C Latency chart

Measures speed, not whether the prompt is better.

D Release, then ask users

Tests on customers what should be tested before them.

THE SIGNAL

Before releasing, is it better, and 300 questions with expected answers. Every signal says evaluation.

> An engineer registers the Correctness judge as a production monitoring scorer on a live assistant.
>
> It produces no useful assessments on production traces.

What should the engineer do instead?

A Increase the sample rate so more traces are scored

B Switch the judge to a larger and more capable evaluator model

C Use judges that need no expected answers, such as Guidelines

D Move the scorer to a separate MLflow experiment

◆ PAUSE AND DECIDE

What should the engineer do instead?

A Higher sample rate

More traces, still no expected answers to compare with.

B Larger evaluator

No model can mark without an answer key.

C Judges needing no expected answers

Guidelines, Safety and relevance work on live traffic.

D Separate experiment

Moves the problem, does not solve it.

THE SIGNAL

Correctness on production traces. Correctness needs expected facts, and live traffic has none.

> A team tests four chunk sizes. 256 tokens: sufficiency 68%, 4.1 million records. 512: 81%, 2.2 million. 1,024: 83%, 1.2 million. 2,048: 74%, 0.6 million.
>
> The requirement is at least 80% RetrievalSufficiency, with the fewest records.

Which chunk size should the team choose?

A. 256 tokens, because small chunks are the most precise

B. 1,024 tokens, because it has the highest recall

C. 2,048 tokens, because it has the fewest records

D. 512 tokens, because it is the first size that passes

◆ PAUSE AND DECIDE

Which chunk size should the team choose?

A 256 tokens

68% fails the threshold, and has the most records.

B 1,024 tokens

Passes at 83%, with fewer records than 512.

C 2,048 tokens

Fewest records, but 74% fails the threshold.

D 512 tokens

Passes, but nearly twice the records of 1,024.

THE SIGNAL

Threshold first, then fewest records. Only 512 and 1,024 pass, and 1,024 is smaller.

> Production monitoring shows the Guidelines judge failing more often, mainly on questions about a newly launched billing plan.
>
> The team wants to fix it without breaking anything else.

What is the best next step?

A Raise every scorer's sample rate to one hundred percent

B Replace the model with a much larger one straight away

C Turn off the Guidelines scorer to reduce the alert noise

D Add failing traces to the eval dataset, fix, then re-evaluate

◆ PAUSE AND DECIDE

What is the best next step?

A Sample everything

More cost, and the problem is already visible.

B Larger model at once

Untested change; the cause is likely missing billing content.

C Turn the scorer off

Hides the problem instead of fixing it.

D Traces into eval set, fix, re-evaluate

Closes the loop: the fix is tested on the real failures.

THE SIGNAL

Monitoring found it; evaluation proves the fix. Failing production traces become evaluation rows.

What to carry forward

- **Evaluation**: before release, fixed dataset, expected answers. **Monitoring**: after, sampled live traffic.
- Monitoring: register a scorer, start it with a **sample rate**. Same scorers as evaluation.
- **Correctness and RetrievalSufficiency cannot monitor**: live traffic has no answer key.
- Production failures become new evaluation rows. The loop closes.
- Choose chunking by evidence: **threshold first, then the cheapest that passes**.

> ON THE EXAM
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> That completes section 3. Next, section 4: module 21 takes AI Search apart, piece by piece.